

Sprint 0 – Phase B: Azure Infrastructure Deployment Runbook

Project: CyberGuard AI (CloudSecure Solutions Ltd) **Scope:** Dev → Staging → Prod scaffolding **Prepared as:** Lead Cloud Engineer / Azure Solutions Architect review

0. Pre-flight: Fix Required Before Any Deployment

`main.bicep` references modules at relative paths like `modules/keyvault.bicep`, but the uploaded files are flat. Required folder layout:

```
infra/  
  main.bicep  
  dev.bicepparam  
  staging.bicepparam      (to be created)  
  prod.bicepparam        (to be created)  
  modules/  
    keyvault.bicep  
    cosmosdb.bicep  
    redis.bicep  
    servicebus.bicep  
    openai.bicep  
    aisearch.bicep  
    storage.bicep  
    appconfig.bicep  
    appservice.bicep  
    budgets.bicep  
    roles.bicep
```

Run locally before anything else:

```
mkdir -p infra/modules  
mv main.bicep dev.bicepparam infra/  
mv keyvault.bicep cosmosdb.bicep redis.bicep servicebus.bicep openai.bicep \  
  aisearch.bicep storage.bicep appconfig.bicep appservice.bicep budgets.bicep \  
  roles.bicep infra/modules/  
  
cd infra  
az bicep build --file main.bicep    # compile-check before touching Azure
```

1. Well-Architected Framework Review

Blocking / Must-Fix Before Production Use

#	File	Issue	Pillar
1	appservice.bicep	No <code>httpsOnly: true, no minTlsVersion, no ftpsState: 'Disabled'</code> on the Web App <code>siteConfig</code>	Security
2	redis.bicep	No Entra ID (managed identity) auth configured. Without it, the app needs a Redis access key — which must live in Key Vault, but nothing currently writes it there or wires the app to read it	Security
3	(none)	No Log Analytics workspace / Application Insights / diagnostic settings anywhere in the template set	Operational Excellence
4	keyvault.bicep	<code>enablePurgeProtection</code> not set (defaults false) — required for Staging/Prod so a vault can't be permanently deleted during the retention window	Reliability/Security
5	(roles/network)	Every resource has public network access enabled by default (OpenAI explicitly <code>Enabled</code> , Storage/Cosmos/Key Vault/Search have no network ACLs or private endpoints) — acceptable for Dev, not for Prod scaffolding	Security
6	(params)	Only <code>dev.bicepparam</code> exists — no <code>staging.bicepparam</code> / <code>prod.bicepparam</code>	Ops

Recommended Improvements

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#	File	Recommendation
7	appconfig.bicep	<code>free</code> SKU has no SLA and no geo-replication — fine for Dev, switch to <code>standard</code> for Staging/Prod
8	budgets.bicep	<code>timePeriod.startDate</code> is hardcoded to <code>2026-07-01</code> — parameterize so the template stays valid after that date passes
9	(resource locks)	No <code>Microsoft.Authorization/locks (CanNotDelete)</code> on Prod resources
10	storage.bicep	No blob versioning / soft-delete policy on <code>blobServices</code> (currently only account-level TLS/HTTPS enforced)
11	openai.bicep	Model <code>gpt-chat-latest</code> version <code>2026-05-05</code> is non-standard naming — confirm exact model/version string and regional availability via <code>az cognitiveservices account list-models</code> before deploying; mismatches fail at deploy time, not plan time
12	(App Service)	No Application Insights resource wired into <code>appservice.bicep</code> app settings (<code>APPLICATIONINSIGHTS_CONNECTION_STRING</code>)
13	appservice.bicep	Plan is <code>B1</code> (no autoscale, no zone redundancy) — fine for Dev; flag for Staging/Prod sizing decision

✅ Done Well

- Cosmos DB: continuous backup (`Continuous30Days`) for PITR is correctly configured.
- Key Vault: `enableRbacAuthorization: true` (no legacy access policies) — correct modern pattern.
- App Service: `SystemAssigned` managed identity is wired through to `roles.bicep` for RBAC — correct pattern, just needs the gaps above closed.
- Storage: `supportsHttpsTrafficOnly` + `minimumTlsVersion: TLS1_2` set correctly.
- Redis: `minimumTlsVersion: 1.2`, `enableNonSslPort: false` — good baseline even without AAD auth.
- Naming via `uniqueString(subscription().id)` avoids global name collisions.
- `roles.bicep` correctly uses **least-privilege built-in roles** (Key Vault Secrets User, not Key Vault Administrator; Search Index Data Contributor, not Service Contributor).

Recommendation: Items 1–6 should be patched before we provision anything beyond Dev. For Dev specifically, items 1, 2, and 3 are still worth fixing now since Dev infra is usually cloned for Staging — cheaper to fix once.

Let me know if you want me to draft the patches for items 1–6 now, or proceed with Dev deployment as-is and patch before Staging.

2. Staged Deployment Plan (Dev first)

Each stage deploys, then validates, before the next stage proceeds. Run from inside `infra/`.

Stage 0 — Auth & Subscription Context

```
az login
az account set --subscription "<SUBSCRIPTION_ID>"
az account show --output table
```

Validate: confirm correct subscription/tenant before anything else.

Stage 1 — What-If (dry run, no resources created)

```
az deployment sub what-if \
  --location westeurope \
  --template-file main.bicep \
  --parameters dev.bicepparam
```

Validate: review the full resource diff. Stop here if anything unexpected appears (wrong RG name, unexpected deletes, etc).

Stage 2 — Resource Group + Key Vault

Deploy just the foundation first (comment out modules 3–12 in `main.bicep` temporarily, or use `--what-if` plus targeted `az deployment group` calls once the RG exists). Simplest path: deploy full template but watch Key Vault complete first since it has no dependencies.

```
az deployment sub create \
  --name "csec-dev-phaseB-$(date +%Y%m%d%H%M)" \
```

```
--location westeurope \  
--template-file main.bicep \  
--parameters dev.bicepparam
```

Validate Key Vault:

```
az keyvault show --name <kvName> --query "properties.{rbac:enableRbacAuthorizatio  
az keyvault network-rule list --name <kvName>
```

Stage 3 — Data Layer (Cosmos DB, Redis, Storage)

These deploy in parallel as part of the same `main.bicep` run above (Bicep handles dependency ordering automatically via output references). Validate each individually:

```
# Cosmos  
az cosmosdb show --name <cosmosName> --resource-group cs-dev-rg \  
  --query "{endpoint:documentEndpoint,backup:backupPolicy.type}" -o table  
az cosmosdb sql container list --account-name <cosmosName> --resource-group cs-de  
  --database-name cloudsecure_platform -o table  
  
# Redis  
az redis show --name <redisName> --resource-group cs-dev-rg \  
  --query "{host:hostName,sslPort:sslPort,provisioningState:provisioningState}" -  
  
# Storage  
az storage account show --name <storageName> --resource-group cs-dev-rg \  
  --query "{tls:minimumTlsVersion,https:supportsHttpsTrafficOnly}" -o table  
az storage container list --account-name <storageName> --auth-mode login -o table
```

Stage 4 — Messaging & AI Services (Service Bus, OpenAI, AI Search)

```
# Service Bus  
az servicebus namespace show --name <sbName> --resource-group cs-dev-rg -o table  
az servicebus queue list --namespace-name <sbName> --resource-group cs-dev-rg -o  
az servicebus topic list --namespace-name <sbName> --resource-group cs-dev-rg -o  
  
# OpenAI  
az cognitiveservices account show --name <openaiName> --resource-group cs-dev-rg  
  --query "{endpoint:properties.endpoint,status:properties.provisioningState}" -o  
az cognitiveservices account deployment list --name <openaiName> --resource-group
```

```
# AI Search
az search service show --name <aisearchName> --resource-group cs-dev-rg \
  --query "{status:status,sku:sku.name}" -o table
```

⚠️ Known risk: flagged item #11 above — if the OpenAI model/version string is wrong for West Europe, this stage fails here. Have a fallback model version ready (check `az cognitiveservices account list-models --location westeurope` first).

Stage 5 — App Configuration + App Service

```
az appconfig show --name <appconfigName> --resource-group cs-dev-rg -o table

az webapp show --name <appserviceName> --resource-group cs-dev-rg \
  --query "{state:state,identity:identity.type,httpsOnly:httpsOnly}" -o table

# Confirm managed identity principal exists
az webapp identity show --name <appserviceName> --resource-group cs-dev-rg
```

Validate connectivity (App Service → dependencies):

```
az webapp config appsettings list --name <appserviceName> --resource-group cs-dev
curl -I https://<appserviceName>.azurewebsites.net # expect a response, even if
```

Stage 6 — RBAC Role Assignments

```
az role assignment list --assignee <principalId> --all -o table
```

Validate: confirm all 7 expected role assignments are present (Cosmos SQL Contributor, Key Vault Secrets User, Storage Blob Data Contributor, Service Bus Data Owner, Cognitive Services User, Search Index Data Contributor, App Configuration Data Reader).

Manual action required: Redis access key still needs to be added to Key Vault manually since AAD auth isn't wired (flagged item #2):

```
REDIS_KEY=$(az redis list-keys --name <redisName> --resource-group cs-dev-rg --qu
az keyvault secret set --vault-name <kvName> --name "redis-primary-key" --value "
```

Stage 7 — Budgets (FinOps)

```
az consumption budget show --budget-name cs-dev-budget --resource-group cs-dev-rg
```

Validate: confirm 80%/100% alert thresholds and contact email are correct.

3. Deployment Summary Template (fill in after each milestone)

```
## Deployment Summary - [Environment] - [Date]

### Resources Deployed
| Resource | Name | Status | Region |
|---|---|---|---|

### Managed Identity Coverage
| Resource | Identity-based access? | Notes |
|---|---|---|

### Secrets in Key Vault (names only, never values)
| Secret Name | Source | Rotation owner |
|---|---|---|

### Connectivity Validation Results
| Stage | Check | Result |
|---|---|---|

### Open Risks / Follow-ups
-

### Sign-off
- Deployed by:
- Validated by:
- Approved to proceed to next stage: Y/N
```

4. Next Steps

1. Confirm whether to patch items 1–6 now (httpsOnly/TLS, Redis AAD or KV-secret wiring, Log Analytics, purge protection, network restrictions, staging/prod param files) before first deploy, or deploy Dev as-is and patch before Staging.
2. Reorganize files into `infra/modules/` structure per Section 0.

3. Run Stage 0 → Stage 1 (`what-if`) and share the output before any `create` runs.